

Exam 1 - Part A

⚠ This is a preview of the draft version of the quiz

The Quiz is Part A (the first of three parts) of Exam 1.

Once you begin this quiz, you will have 20 minutes to complete it. The 20 minutes begins when you start the quiz, so read the instructions before you start. These instructions are the same for all parts of the exam.

Be sure to complete **all** parts of the exam before 9:00pm (Madison time).

Please remember that during the exam window, you are not permitted to communicate with other students in class. Note: if someone breaks this rule (e.g., by sending a message or posting

- Quiz Type

Graded Quiz
- Points

31
- Assignment Group

Exams
- Shuffle Answers

No
- Time Limit

20 Minutes
- Multiple Attempts

No
- View Responses

Always
- Show Correct Answers

Immediately
- One Question at a Time

No

Due	For	Available from	Until
Mar 8 at 9pm	Everyone	Mar 8 at 5:30pm	Mar 8 at 9pm

Preview

Score for this quiz: 0 out of 31

Submitted Mar 8 at 1:43pm

This attempt took less than 1 minute.

Unanswered

Question 1

0 / 3 pts

What color is the square that is farthest to the **right**? Note: squares might be drawn on top of one another.

We are asking about visible squares.

```
1 context.translate(200,200);
2 context.strokeStyle = "black";
3 for(let color of ["red", "green", "blue", "purple" ]) {
4     context.strokeRect(0,0, 10,10);
5     context.translate(20,0);
6     context.strokeStyle = color;
7 }
```

☐ red

☐ green

☐ blue

☐ purple

☐ black

Correct Answer

This draws 4 squares. In the loop, the color is changed after the square is drawn, so the squares are black, red, green and blue. In this case each square is shifted to the right, so all 4 squares show in a row.

Unanswered

Question 2

0 / 3 pts

How many circles will be visible in this SVG?

```
1 <svg xmlns="http://www.w3.org/2000/svg"
2     xmlns:xlink="http://www.w3.org/1999/xlink"
3     width="200px" height="200px">
4
5 <def>
6 <circle id="c1" cx="40" cy="20" r="20" />
7 </def>
8 <use xlink:href="#c1" transform="translate(40,90)" />
9 <use xlink:href="#c1" transform="translate(90,90)" />
10
11 </svg>
```

Note: if a circle is drawn on top of another circle, we do not consider the covered circle visible.

You Answered

Correct Answers

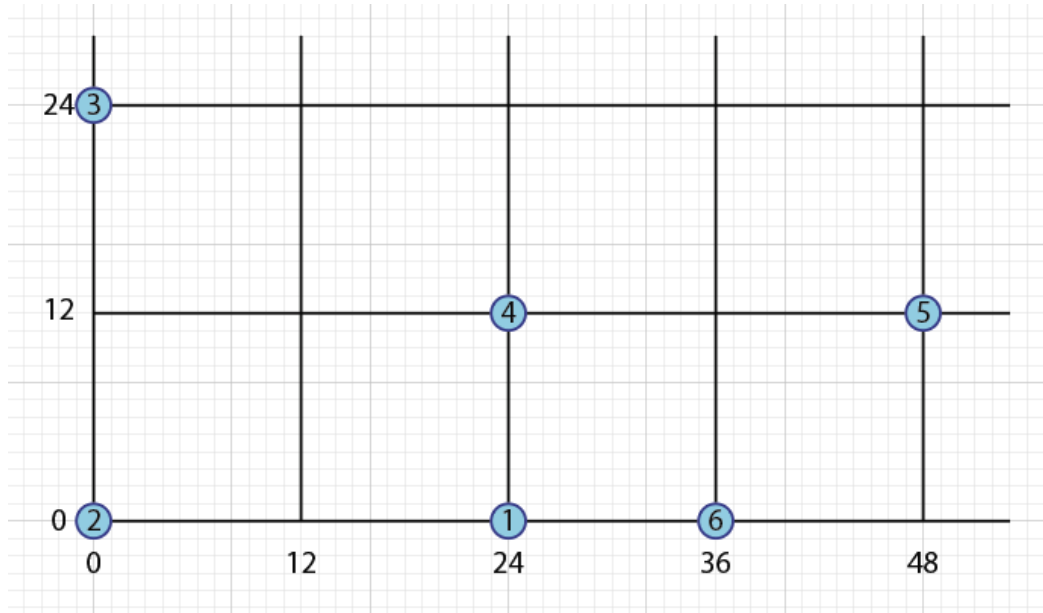
2 (with margin: 0)

The circle inside of the "def" is not drawn, so only the two uses of the def get drawn.

The next few questions refer to this figure:

A looped Catmull-Rom (cardinal spline with $s=1/2$ or tension=0) spline is drawn through these 6 points (point 1 is at the bottom center at (24,0)). The points go clockwise, and point 1 follows point 6.

Warning: this coordinate system is drawn with Y up (so Y=0 is at the bottom).



Unanswered

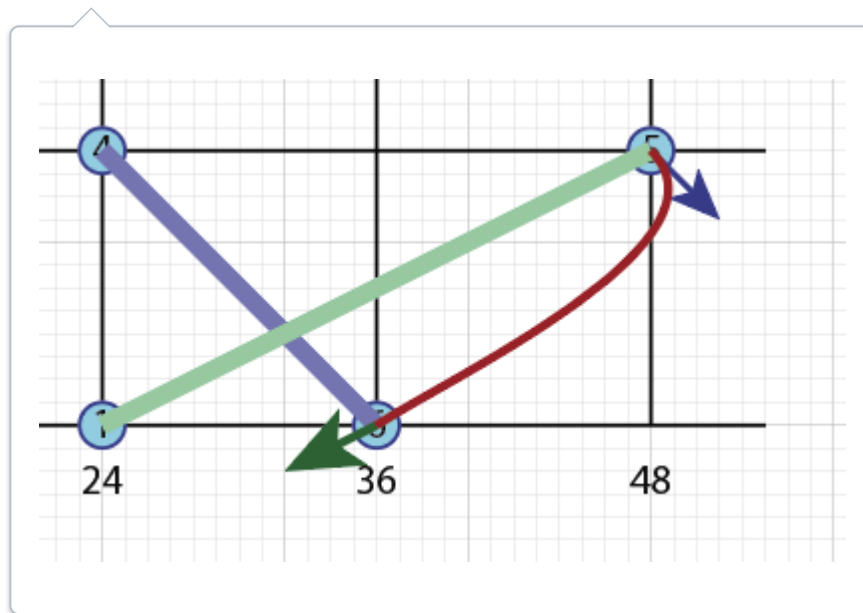
Question 3

0 / 3 pts

Does the curve cross the line $x=48$?
☐ The curve touches $x=48$, but never crosses it (x is never > 48)

Correct Answer

☐ The curve crosses $x=48$ twice, both crossings are at or below $y=12$
☐ The curve crosses $x=48$ twice, one crossing is above $y=12$, the other is below $y=12$



Unanswered

Question 4

0 / 6 pts

If we convert the curve to a series of 6 Bezier segments, the segment that connects point 3 to point 4 has what 4 control points?

(we'll give you the first one)

(0,24), (,), (,), (,)

Answer 1:

Unanswered

(You left this blank)

Correct Answer

4

Answer 2:

Unanswered

(You left this blank)

Correct Answer

26

Answer 3:

You Answered

(You left this blank)

Correct Answer

16

Answer 4:

You Answered

(You left this blank)

Correct Answer

14

Answer 5:

You Answered

(You left this blank)

Correct Answer

24

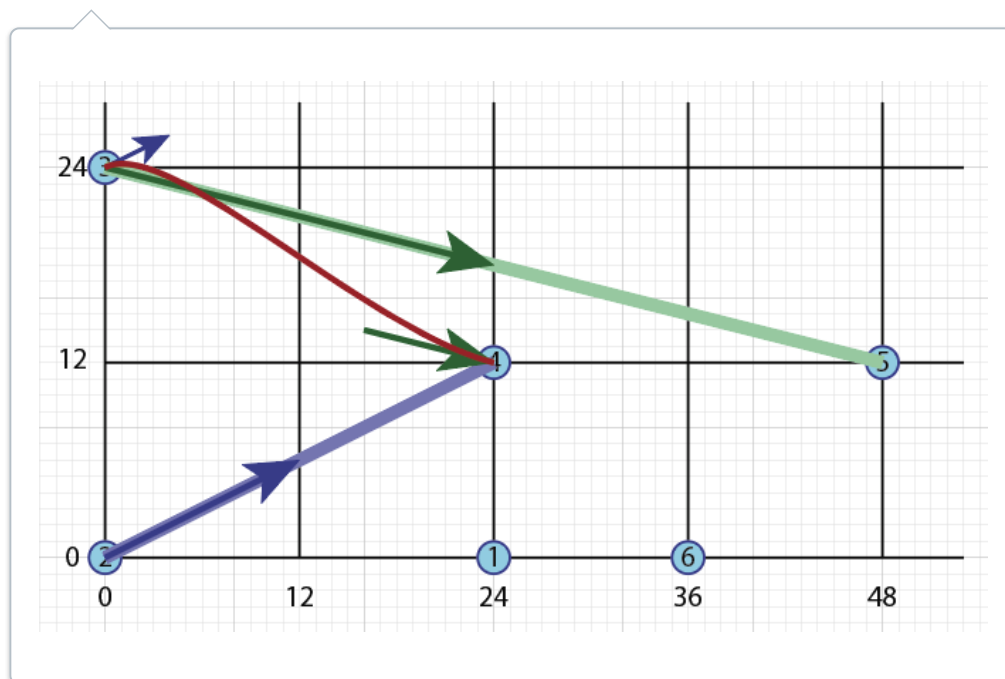
Answer 6:

You Answered

(You left this blank)

Correct Answer

12



Unanswered

Question 5

0 / 3 pts

The tangent to the curve at point 1 depends on which points?

☐ 1

Correct Answer

☐ 2☐ 3☐ 4☐ 5

Correct Answer

☐ 6

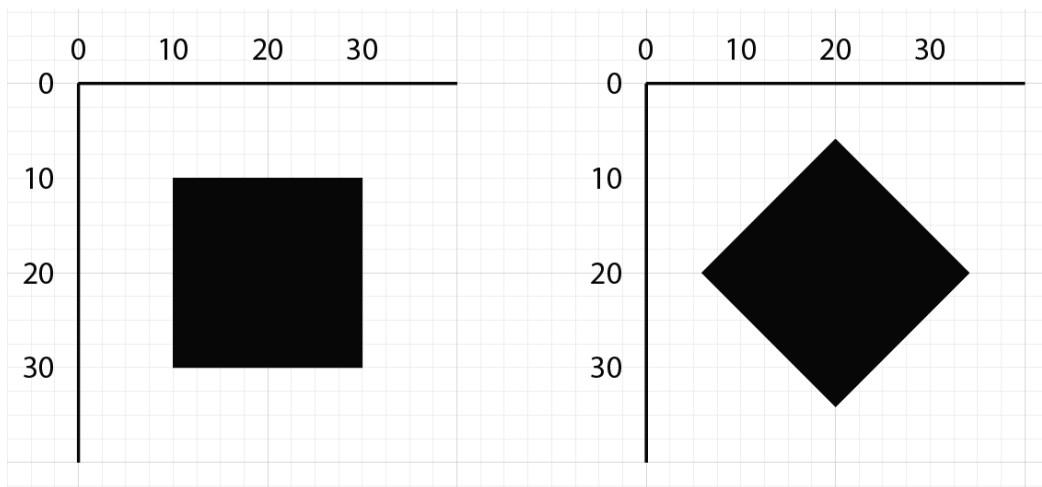
Unanswered

Question 6

0 / 3 pts

We have implemented a rectangle function for the HTML5 2D Canvas (what we use in class) that takes the two corners as arguments and draws a rectangle in the current coordinate system.

So, "rectangle(10,10,30,30);" produces the square in the left of these two images.



Unlike the built in rectangle commands, the arguments to this function are point positions (x1,y1,x2,y2), and the rectangle is properly transformed.

Suppose we want to draw the picture on the right (which has the square rotated by 45 degrees about its center).

And we need to write the program by using the following 6 lines in some order:

```
/* 1 */ rectangle(10,10,30,30) ;  
/* 2 */ context.rotate(Math.PI/4); /* this is 45 degrees converted to radians */  
/* 3 */ context.translate(20,20);  
/* 4 */ context.translate(-20,-20);  
/* 5 */ context.translate(10,10);  
/* 6 */ context.translate(-10,-10);
```

Which of the following is true?

☐

There is a solution that is fewer than 4 lines. (using a combination of 1-6)

☐

The solution of lines 3,2,4,1 (in that order) is correct

☐

The solution of lines 4,2,3,1 (in that order) is correct

☐

The solution of lines 5,2,6,1 (in that order) is correct

☐

The solution of lines 6,2,5,1 (in that order) is correct

☐

The solution of lines 2,3,4,1 (in that order) is correct

☐

The solution of lines 1,3,2,4 (in that order) is correct

Correct Answer

3,2,4,1 is the right answer - move the center of the coordinate system to the center of the object, rotate it, move it back

there is a solution that does a single rotate and a single translate - but it requires translating by an amount that is not a multiple of 10

there are solutions that use lines 5 and/or 6 multiple times (but those are longer than 4 steps)

Unanswered

Question 7

0 / 6 pts

The commands

```
context.translate(1,2);
```

```
context.translate(3,4);
```

```
context.translate(4,5);
```

would be the same as the single statement:

```
context.translate( , );
```

The commands

```
context.scale(1,2);
```

```
context.scale(3,4);
```

```
context.scale(4,5);
```

would be the same as the single statement:

```
context.scale( , );
```

And the commands

```
context.rotate(1);
```

```
context.rotate(3);
```

```
context.rotate(4);
```

would be the same as the single statement:

```
context.rotate(  );
```

Answer 1:

ou Answered

(You left this blank)

orrect Answer

8

Answer 2:

ou Answered

(You left this blank)

orrect Answer

11

Answer 3:

ou Answered

(You left this blank)

orrect Answer

12

Answer 4:

ou Answered

(You left this blank)

orrect Answer

40

Answer 5:

ou Answered

(You left this blank)

orrect Answer

8

Inanswered

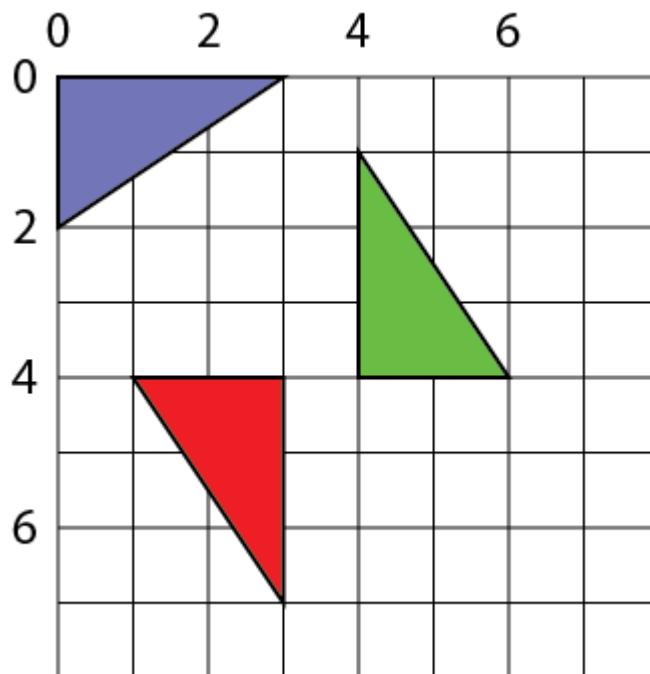
Question 8

0 / 4 pts

This program:

```
context.fillStyle = "blue";  
drawTriangle();  
context.save();  
context.rotate(-Math.PI/2);  
context.translate(gx,gy);  
context.fillStyle = "green";  
drawTriangle();  
context.restore();  
context.save();  
context.translate(rx,ry);  
context.rotate(Math.PI/2);  
context.fillStyle = "red";  
drawTriangle();  
context.restore();
```

Draws this picture:



What are the values of the constants?

```
const gx =  ;  
const gy =  ;  
const rx =  ;  
const ry =  ;
```

Answer 1:

You Answered

(You left this blank)

Correct Answer

-4

Answer 2:

You Answered

(You left this blank)

Correct Answer

4

Answer 3:

You Answered

(You left this blank)

Correct Answer

3

Answer 4:

You Answered

(You left this blank)

Correct Answer

4

Quiz Score: 0 out of 31